Endpoint-fiber EndRing PKE instantiation: exact transport obstruction

GOAL-SSI-2fa82a · BATCH-0cfd90 · zero-run research report

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Result at a glance

Outcome — independently reviewed: this round did **not** produce a public-key encryption scheme. It established one narrower algebraic diagnostic for the attempted CEFC-I key law: transporting a complete base endomorphism ring along a supported cyclic isogeny by integral pushforward produces a strict suborder of exact index d^3, while normalized conjugation is not uniformly integral. The next KeyGen call, `CompleteBasisAtEndpoint`, has no supplied algorithm, output law, verification method, polynomial cost, or failure law. The attempt therefore stops at O1.

All six construction obligations remain open. IND-CPA, active security, attacker-to-complete-EndRing extraction, novelty, QPT/QROM security, and NIST Category 5 are separately unestablished. There were no scientific, implementation, formalizer, experiment, benchmark, or parameter-selection runs.

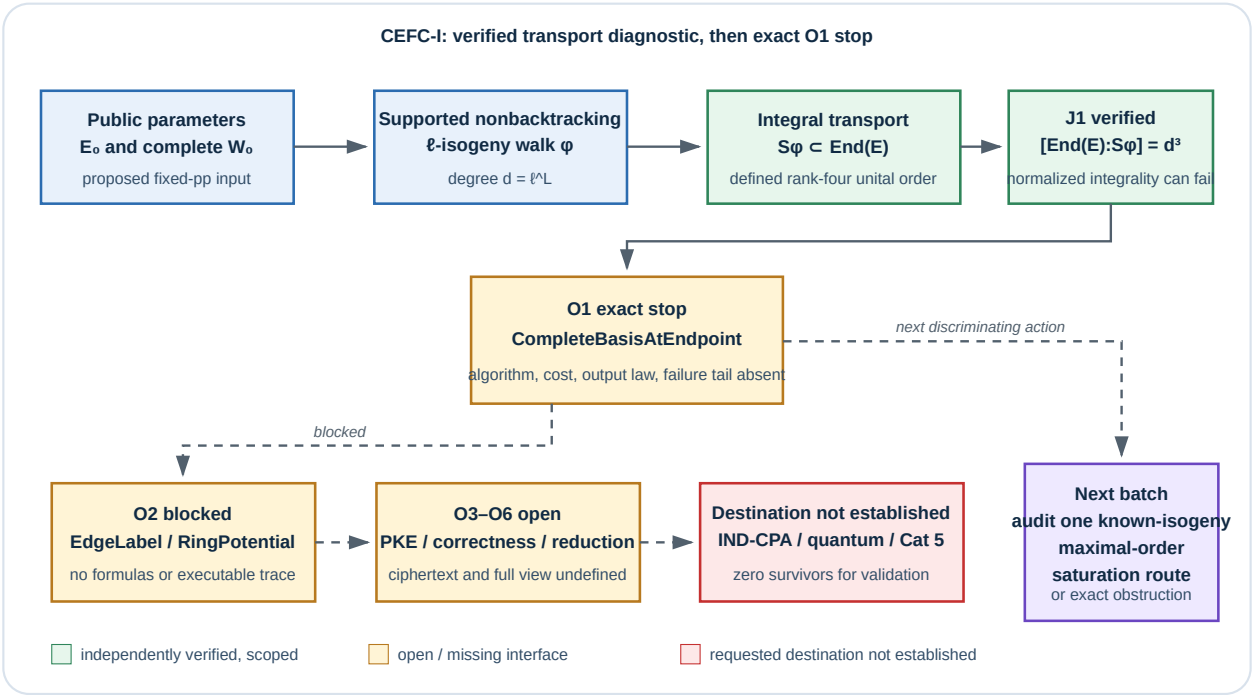


Figure 1: CEFC-I evidence flow. Green is independently verified within the stated scope, amber is a missing interface or open obligation, and red means the requested destination was not established.

Question and search scope

The goal asks for a novel supersingular-EndRing PKE that is quantum secure and satisfies a NIST Category-5 target. This batch tested exactly one mechanism, CEFC-I:

1. start from public parameters containing a supersingular curve E0 over the quadratic field and a complete evaluable saturated basis W0 of End(E0);
2. sample a length-L nonbacktracking chain of separable degree-ell isogenies with composite phi and degree $d = ell^L$;
3. derive a complete evaluable saturated basis WE of the endpoint ring;
4. use one public, nontrivial, zero-circulation `EdgeLabel` and a polynomial complete-ring `RingPotential` to obtain endpoint-fiber agreement;
5. build and evaluate a PKE, then separately discharge IND-CPA, active security, complete-EndRing extraction, novelty, quantum-model, and Category-5 gates.

The frozen methodology was "propose, evaluate, then stop at the earliest exact obstruction." It prohibited replacing PKE with a KEM, keeping the walk or orientation as undeclared secret advice, treating full graph/Hom enumeration as polynomial, or asserting security from recipient functionality alone.

The attempted key law

Proposed: for valid fixed public parameters, the first isogeny edge is uniform among the $ell + 1$ actual cyclic subgroups; each later edge is uniform among the ell actual outgoing edges other than the preceding dual. Parallel actual edges retain multiplicity. The endpoint distribution is the exact pushforward of these coin strings, not a uniform-curve distribution.

Let phi map E0 to E as the canonically transported composite, with degree d, and let

$$W_0 = (1, \alpha_2, \alpha_3, \alpha_4)$$

be a complete basis of End(E0). The available integral maps are

$$F_\phi(\alpha) = \phi\alpha\widehat{\phi}, \qquad S_\phi = \mathbb{Z} \cdot 1 + \sum_{i=2}^{4} \mathbb{Z} \cdot F_\phi(\alpha_i).$$

Derived by the producer and independently verified by J1: S-phi is a rank-four unital order. Writing tau(alpha) for rational conjugation by phi, one has F-phi(alpha) = d times tau(alpha). Relative to the conjugate basis

$$(1, \tau(\alpha_2), \tau(\alpha_3), \tau(\alpha_4)),$$

the change is diagonal with entries $(1, d, d, d)$. Because the conjugate of End(E0) and End(E) are maximal orders of equal covolume, and S-phi is contained in End(E),

$$[\mathrm{End}(E) : S_\phi] = d^3 > 1.$$

The normalized conjugates are not uniformly integral: integrality would require alpha to preserve the kernel of phi, whereas the complete ell-adic endomorphism order acts as the full matrix algebra and does not preserve a fixed kernel line for every alpha.

This establishes that two obvious substitutions fail:

- returning the transported four-element tuple returns only an index-d^3 suborder;
- dividing every transported element by d does not uniformly return integral endomorphisms.

It does **not** establish a lower bound or the nonexistence of a polynomial saturation algorithm.

Exact stopping point

The next honest pseudocode line would be:

```
W_E <- CompleteBasisAtEndpoint(pp, phi, W0)
```

Open: no implementation, evaluable output representation, complete-ring certificate, conversion method, exact cost, retry law, or failure bound is supplied. KeyGen therefore does not complete. `EdgeLabel`, `RingPotential`, `Encrypt`, and `Decrypt` are literal null interfaces in the blind public statement, so no executable PKE trace follows.

The acceptance criterion for a successor is precise: for every declared valid parameter tuple and supported walk, give one uniform polynomial-bit-time and polynomial-memory algorithm that outputs a complete evaluable saturated basis of End(E), with full-ring verification, exact output representation, charged preprocessing/conversion, and a success/retry tail. The output must not hide retained path advice.

Independent review composition

The review plan was frozen before review. J1, J2, and J3 ran in mutually blind sessions; J3 rederived from the literal public statement without reading producer proofs. The repository checker reported:

```
PASS: 3 report(s), every joint owned and attested, blindness respected, controls declared
```

This is a procedure check, not a mathematical or security proof.

Joint	Sole owner	Verdict	Scoped result
J1: O1/O2	`TASK-20261014-604885`	holds	The narrow integral-transport obstruction, exact d^3 index, and nonuniform normalized integrality hold. O1/O2 remain open; polynomial completion is not ruled out.
J2: O3/O4	`TASK-20261014-9ff2ea`	inconclusive	No label, potential, PKE, relation domain, correctness theorem, or full-ring-necessity exclusion exists. O3/O4 remain open.
J3: O5/O6 and separate gates	`TASK-20261014-0754ea`	inconclusive	No real PKE game, extractor, complete adversarial view, novelty evidence, quantum interface, or concrete target exists. O5/O6 and all separate gates remain open.

Requirement-to-evidence table

Requested requirement	Evidence achieved in this batch	Status
Exact real key/walk law	Conditional fixed-parameter walk and endpoint marginal are explicit; Setup family, successful complete-key joint law, and retry law are absent.	open
Complete evaluable endpoint ring	Naïve transport is independently shown to have index d^3; completion/saturation algorithm is absent.	open
Public nontrivial zero-circulation label	No concrete group or label formula.	open
Polynomial complete-ring potential	No concrete potential algorithm or cost.	open
Executable PKE syntax	`Encrypt` and `Decrypt` are undefined.	not established
Correctness	No fiber theorem, message law, failure space, or bound.	open
Recipient asymmetry/full-ring necessity	Public coboundary/base-ring and smaller-witness controls are not excluded.	open
IND-CPA	No complete real PKE experiment or proof.	not established
Active security	No active syntax. Raw endpoint-plus-XOR would be conditionally IND-CCA2-broken by a one-query payload flip, but CEFC-I does not instantiate it.	not established
Attacker-to-complete-EndRing reduction	No challenge embedding, simulator, complete-basis extractor, success/loss, or runtime.	open
Novelty	No completed mechanism or primary-source comparison.	not established
QPT/QROM security	No quantum assumption, query interface, simulator/programming rule, extractor, or loss.	not established
NIST Category 5	No completed scheme, normative comparison, parameters, attack portfolio, reduction loss, measurements, or matched cost evidence.	not established

Controls and security boundaries

The reviews applied controls whose conclusions are known to be false:

- A public coboundary `a(e) = h(head(e)) - h(tail(e))` has perfect circulation and endpoint agreement but makes the value public.
- A zero or constant label fails nontriviality and yields no recipient-only value.
- A path hash does not factor through endpoints without a same-endpoint collision/fiber theorem.
- Calling a full basis does not prove it is necessary: one evaluator, finite torsion action, cache, compressed decoder, or opaque advice could suffice unless excluded or converted to a complete basis.
- The implication "a ring witness enables decryption" runs in the wrong direction for an EndRing-only security reduction. A successful scheme attacker must be transformed into a solver outputting a complete evaluable saturated basis on the supplied challenge curve.
- For a raw pair `(C, u = m XOR z_E(C))` that accepts same-endpoint payload flips, querying `(C, u XOR Delta)` after the challenge recovers `m` with one byte-distinct CCA2 query. This is a conditional control, not an attack on the undefined CEFC-I encryption algorithm.

Disposition and next discriminating action

Independently reviewed composition: CEFC-I has zero survivors for validation. It is not a complete PKE; IND-CPA and active security are not established; no complete-EndRing reduction, novelty result, quantum-security result, or Category-5 result exists.

The most discriminating next round is narrower than another full scheme proposal: investigate exactly one known-isogeny completion route. Starting from W0 and the factored cyclic phi, either provide a uniform polynomial algorithm that reconstructs and evaluates the saturated maximal order End(E), with a full-ring certificate and exact cost/failure law, or identify the earliest exact obstruction. Only after that succeeds should construction of `EdgeLabel` and `RingPotential` resume.

Limitations and canonical-graph status

This was a symbolic zero-run round. It selected no parameter family, measured no performance, retrieved no new primary literature, and proves no universal impossibility. The result applies only to the frozen CEFC-I integral/normalized transport attempt and its null downstream interfaces.

Canonical graphs were checked and left unchanged. The related editable/rendered studies under `research/ssi-endring-pke/` were not historical evidence to mutate, and no canonical graph under `knowledge/frontiers/ecdlp/` depicts this mechanism. This batch establishes no new verified cryptographic edge or complete construction; adding one to a canonical graph would overstate the evidence. The new `flow.mmd`/`flow.svg` pair is therefore a batch-local evidence diagram.

Frozen evidence and provenance

- Producer composition input (`final-composition.yaml`):

  `8dc5a0a5bb4f47d8f4831877adb178f164aa235f6a37ca3c284e9a6eec2fdc55`

- J1 review:

  `e1adcefc2bf4efb260d6ac6e972406fcde5d62f8de37d37a760bdafda965c6f4`

- J2 review:

  `e51a17a22f818a62fb16a240b27081c6094949edbdb873609b73162fc3545f09`

- J3 review:

  `bec42a21cf2752ba03dd0ca534a2cbb1aa56439f24190a076c4923728fd2be14`

Primary internal sources:

- `coordination/design/TASK-20261014-11b3d6/composition/TASK-20261014-efe1bc/final-composition.yaml`

- coordination/design/TASK-20261014-11b3d6/tasks/TASK-20261014-460fea/obstruction.yaml
- coordination/design/TASK-20261014-11b3d6/reviews/TASK-20261014-604885/review.yaml
- coordination/design/TASK-20261014-11b3d6/reviews/TASK-20261014-9ff2ea/review.yaml
- coordination/design/TASK-20261014-11b3d6/reviews/TASK-20261014-0754ea/review.yaml
- coordination/design/TASK-20261014-11b3d6/archives/TASK-20261014-feb4ff/ledger-receipt.json

Claim labels used in this report are proposed, derived, or independently reviewed. Nothing here is measured.